EXAM PACK

EEE459

Compound Semiconductor and Hetero-Junction Devices



Exam Pack

Session 2021–22 *Solutions and formula sheet of Term Test 1 (the final will be added)*

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Prepared by Nur Mohammad

November 2026

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PAST PAPER SOLUTIONS

EEE459

Compound Semiconductor and Hetero-Junction Devices

Question Paper Solutions

Session 2021–22

Term Test 1 (Q7); the final examination will be added as Q1–Q6

QUESTION MAP

Q7

20 marks

EACH ANSWER

Theory → Answer → Keyword Explanation → Notes and Remarks → Final Concise Answer

Prepared by Nur Mohammad

November 2026

BACK OF THE COVER

The paper at a glance

30 min	20
Total time	Total marks

Time = marks × 1.5 min (30 min for 20 marks); a part with its own time (a term test) uses that. Tick the questions you will answer; each sub-part links to its solution.

Q	Sub-part	Marks	Time	Pick
Term Test 1 · 20 marks · 30 min				
Q7	(a) Direct and Indirect Gaps of GaAsP Against Composition	3	4 min	☐
	(b) Density of States from 3D to 0D	8	12 min	
	(c) Electric Field of a Metal–n-Semiconductor Junction	9	14 min	

Formulas and constants follow Neamen, Semiconductor Physics and Devices (4th ed.); compound-semiconductor data follow Sze and Ng, Physics of Semiconductor Devices.

TT1, EEE 459

TT1, EEE 459 **Marks: 20** **Time: 30 minutes**

Term Test 1 (its questions 1 to 3 are 7(a) to 7(c))

[TT1, EEE 459; Marks: 20; Time: 30 minutes]

7. (a) Show the compositional dependence of direct-energy gap Γ and indirect energy gap X for GaPAs. Which one is more stepper [*sic*: should be steeper] and why? 1+2

   (b) Explain how the density of states changes qualitatively as the dimensionality reduces from 3D → 2D → 1D → 0D. Sketch $D(E)$ vs E for each case. 8

   (c) A metal-semiconductor (n-type) junction is formed where the work function of metal is greater than that of the semiconductor. With necessary assumptions, derive expression for electric field in both regions. Also sketch the electric field profile for the both regions. 9

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7. (a) Show the compositional dependence of direct-energy gap Γ and indirect energy gap X for GaPAs. Which one is more stepper [*sic*: should be steeper] and why? [Marks: 1+2]

(b) Explain how the density of states changes qualitatively as the dimensionality reduces from 3D → 2D → 1D → 0D. Sketch $D(E)$ vs E for each case. Marks: 8

(c) A metal-semiconductor (n-type) junction is formed where the work function of metal is greater than that of the semiconductor. With necessary assumptions, derive expression for electric field in both regions. Also sketch the electric field profile for the both regions. Marks: 9

7.(a) Show the compositional dependence of direct-energy gap Γ and indirect energy gap X for GaPAs. Which one is more stepper [*sic*: should be steeper] **and why? [Marks: 1+2]**

1. Theory

1. What is $GaAs_{1-x}P_x$? (Core Idea)

$GaAs_{1-x}P_x$ (the paper writes "GaPAs") is a **ternary alloy**. Its crystal is the zinc-blende lattice of GaAs, but on the group-V sites a fraction x of the atoms are phosphorus (P) and the rest, $1 - x$, are arsenic (As). The fraction x is the **mole fraction** (or composition):

- $x = 0$: pure GaAs.
- $x = 1$: pure GaP.
- In between: an alloy whose properties change smoothly from those of GaAs to those of GaP.

We make such alloys to **tune the band gap**, and with it the colour of the light an LED gives ($\lambda = hc/E_g$, or $\lambda(\mu\text{m}) = 1.24/E_g(\text{eV})$).

2. Why does a semiconductor have more than one "gap"?

An electron in a crystal has an energy E that depends on its **wave vector** k (its crystal momentum divided by $\hbar$). The plot of E against k is the **E–k diagram**. The conduction band of a III–V compound is not one simple parabola; it has several low points, called **valleys**:

- the Γ **valley** at $k = 0$, the centre of the Brillouin zone;
- the **X valleys** at the edge of the zone along the [100] directions;
- the L valleys along [111] (they do not matter here).

The top of the valence band is always at $k = 0$ (Γ). So there are two gaps to compare:

- the **direct gap** E_Γ: from the valence-band top to the Γ valley, straight up in k;
- the **indirect gap** E_X: from the valence-band top to the X valleys, with a change of k.

The band gap of the material is the **smaller** of the two. If $E_\Gamma < E_X$ the material is **direct** (GaAs); if $E_X < E_\Gamma$ it is **indirect** (GaP, Si).

3. Why does direct or indirect matter for light?

A photon carries almost no momentum. An electron at the bottom of the Γ valley sits right above the holes at the valence-band top (same k), so it can drop and emit a photon on its own: fast, efficient light. An electron in an X valley has a large k; to fall to $k = 0$ it must also give its momentum to a **phonon** (a lattice vibration). Such three-body events are rare, so indirect materials emit light very weakly. That is why the direct–indirect crossover is the most important feature of the GaAsP band diagram.

4. How do the gaps change with x?

Both gaps grow as P replaces As, because P is a smaller atom: the lattice constant shrinks from 5.653 Å (GaAs) to 5.451 Å (GaP), the bonds become shorter and stronger, and the bonding and antibonding levels move further apart. The measured dependence is close to a straight line with a small upward bow (Sze and Ng; Casey and Panish):

$$E_\Gamma(x) = 1.424 + 1.150x + 0.176x^2 \text{ eV}$$

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$$E_X(x) = 1.900 + 0.144x + 0.211x^2 \text{ eV}$$

where the constant term is the GaAs value, the linear term is the main trend and the x^2 term is the **bowing** (the alloy's small departure from a straight line). At $x = 1$ these give GaP: $E_\Gamma = 2.75$ eV and $E_X = 2.26$ eV.

5. Why is the Γ gap so much more sensitive?

The states at the bottom of the Γ valley are **s-like antibonding** states: they are built from the outer s orbitals of the atoms, which sit right on the atoms and overlap strongly with their neighbours. When the bond gets shorter (P instead of As), this overlap grows sharply and the antibonding s level is pushed up a lot. In addition, the s level of the P atom lies deeper than that of As, which raises the gap further. The strength of this effect is measured by the **deformation potential**: about −8 eV for the Γ gap of GaAs, but only about +1 to +2 eV for the X gap (Sze and Ng).

The states at the X point are a mixture with much **p-like** character, spread between the atoms. Their energy hardly depends on the bond length, so E_X rises only slowly.

2. Answer

Compositional dependence. In $GaAs_{1-x}P_x$ both gaps increase with the P fraction x, nearly linearly with a small bowing (Sze and Ng):

$$E_\Gamma(x) = 1.424 + 1.150x + 0.176x^2 \text{ eV}, \qquad E_X(x) = 1.900 + 0.144x + 0.211x^2 \text{ eV}$$

x	0 (GaAs)	0.2	0.4	0.48	0.6	0.8	1 (GaP)
E_Γ (eV)	1.424	1.661	1.912	2.018	2.177	2.457	2.750
E_X (eV)	1.900	1.937	1.991	2.018	2.062	2.150	2.255

Table 1: Summary

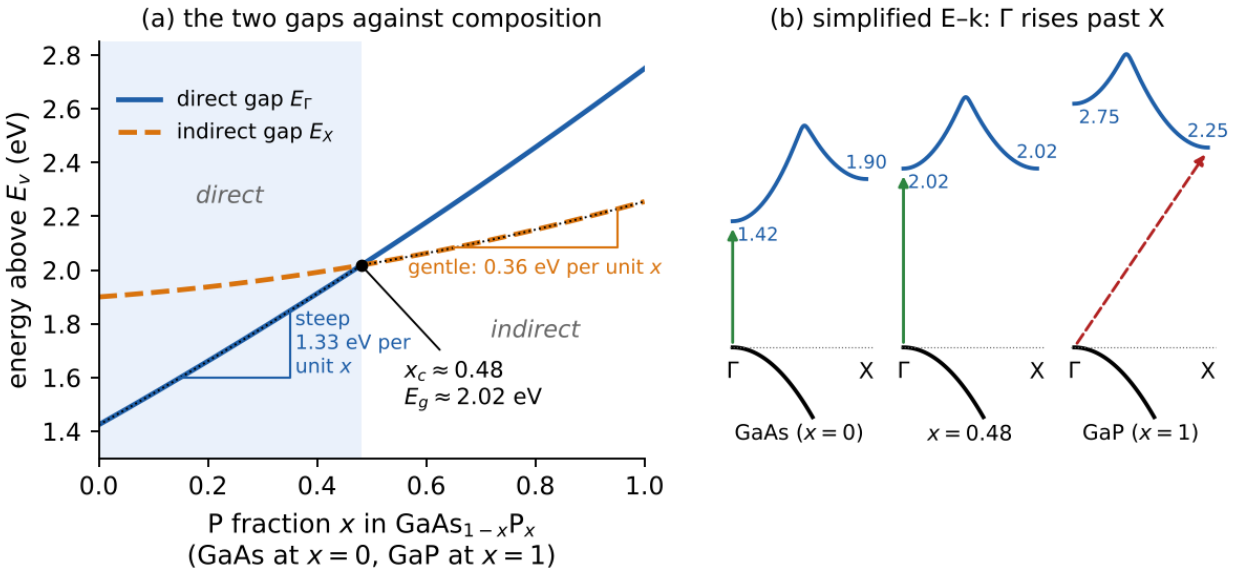


Figure 1: Direct (Γ, solid blue) and indirect (X, dashed orange) gaps of $GaAs_{1-x}P_x$ against x, crossing at x = 0.48; simplified E-k diagrams for GaAs, the crossover and GaP

Reading the figure. In panel (a) the solid blue line is the direct gap and the dashed orange curve the indirect gap; the dotted black line follows the lower of the two, which is the band gap. They cross at $x_c \approx 0.48$, $E_g \approx 2.02$ eV (Neamen quotes $x \approx 0.45$). The shaded part, $x < x_c$, is direct; beyond it the alloy is indirect. The two small triangles show the slopes. Panel (b) gives the simplified E–k diagram (conduction band solid blue, valence band solid black) at three compositions. In GaAs the Γ valley (1.42 eV) is far below X (1.90 eV) and the transition is vertical (solid green arrow). At x_c both valleys

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are at 2.02 eV. In GaP the Γ valley (2.75 eV) is above X (2.25 eV), so the lowest transition is slanted (dashed red arrow) and needs a phonon.

Which is steeper: the **direct (Γ) gap**. From GaAs to GaP it rises by $2.750 - 1.424 = 1.33$ eV, while the indirect (X) gap rises by only $2.255 - 1.900 = 0.36$ eV, a ratio of about 3.7. Its slope $dE_\Gamma/dx = 1.15 + 0.35x$ is 1.15 to 1.50 eV per unit x, against $dE_X/dx = 0.14 + 0.42x$, only 0.14 to 0.57 eV.

Why:

1. The Γ-valley minimum is an **s-like antibonding state**, built from the atomic s orbitals that overlap strongly between neighbours. Replacing As by the smaller P shortens the bond (the lattice constant falls by 3.6 %), the overlap grows and this level rises sharply; the deeper s level of P adds to it. Its deformation potential is large (about −8 eV).
2. The X-valley minimum has mostly **p-like** character and is much less sensitive to bond length and to the anion (small deformation potential), so it rises slowly.
3. Because E_Γ starts lower (1.42 against 1.90 eV) but rises faster, it overtakes E_X at $x_c \approx 0.45$–0.48: GaAsP is direct below and indirect above, which is why efficient GaAsP LEDs stop near 0.63 µm unless an isoelectronic trap (N) is added.

3. Keyword Explanation

1. Mole fraction

The fraction x of one kind of atom on its sublattice: here the fraction of group-V sites that hold P.

2. Bowing parameter

The coefficient of x^2 in a gap-against-composition formula; it measures how much the alloy departs from a straight line between its two end compounds.

3. Deformation potential

The change of a band edge per unit relative change of crystal volume; a large value means the band edge moves a lot when the lattice is squeezed.

4. Notes and Remarks

- Neamen gives the crossover as $x \approx 0.45$ ($E_g \approx 1.98$ eV); the fits above cross at $x = 0.48$, $E_g = 2.02$ eV. Different measurements put it between 0.45 and 0.49.
- The same pattern (fast Γ, slow X) holds in $Al_xGa_{1-x}As$, which becomes indirect near $x = 0.45$, and in the Si–Ge alloy for the L and X valleys.
- This question is also part of 2(c) of the 2020–21 paper, which adds the isoelectronic trap (extra reading).

7.(b) Explain how the density of states changes qualitatively as the dimensionality reduces from 3D → 2D → 1D → 0D. Sketch $D(E)$ vs E for each case. Marks: 8

1. Theory

1. What is the density of states? (Core Idea)

In a crystal an electron can only be in certain **allowed states**. The **density of states** $D(E)$ (Neamen writes $g(E)$) says how many allowed states there are per unit energy, per unit size of the sample, at energy E:

$$\text{number of states between } E \text{ and } E + dE = D(E)\,dE$$

per unit volume in 3D, per unit area in 2D and per unit length in 1D. It matters because the number of electrons at an energy is (number of states) × (probability that a state is occupied, the Fermi function). So $D(E)$ fixes the carrier density, the optical absorption and the gain of a laser.

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2. Energy and wave vector of a free carrier

Near the bottom of the conduction band an electron behaves like a free particle with an **effective mass** m^*. Its energy above the band edge depends on its wave vector $\mathbf{k} = (k_x, k_y, k_z)$ as

$$E = E_c + \frac{\hbar^2}{2m^*}\left(k_x^2 + k_y^2 + k_z^2\right)$$

where $\hbar = h/2\pi$ is the reduced Planck constant. A larger k means a larger kinetic energy.

3. Allowed k values: the grid in k-space

Put the electron in a box of side L. With periodic boundary conditions, the wave $e^{ik_x x}$ must repeat after a length L, so $k_x L = 2\pi n_x$ with n_x an integer:

$$k_x = \frac{2\pi n_x}{L}, \qquad n_x = 0, \pm 1, \pm 2, \ldots$$

and the same for k_y and k_z. The allowed $\mathbf{k}$ form a regular grid with spacing $2\pi/L$. Each grid point holds *two* electrons (spin up and spin down). To find $D(E)$ we count grid points between k and $k + dk$ and turn dk into dE.

4. What confinement does

If the box is made very thin in one direction (a few nanometres), the electron behaves like a wave trapped between two walls: only standing waves fit, and the energy for motion in that direction is **quantised**. For a well of width L_z with very high walls,

$$E_n = \frac{\hbar^2}{2m^*}\left(\frac{n\pi}{L_z}\right)^2, \qquad n = 1, 2, 3, \ldots$$

The electron is still free in the other directions. Each E_n starts a **subband**: a set of states with that frozen energy plus free motion in the remaining directions.

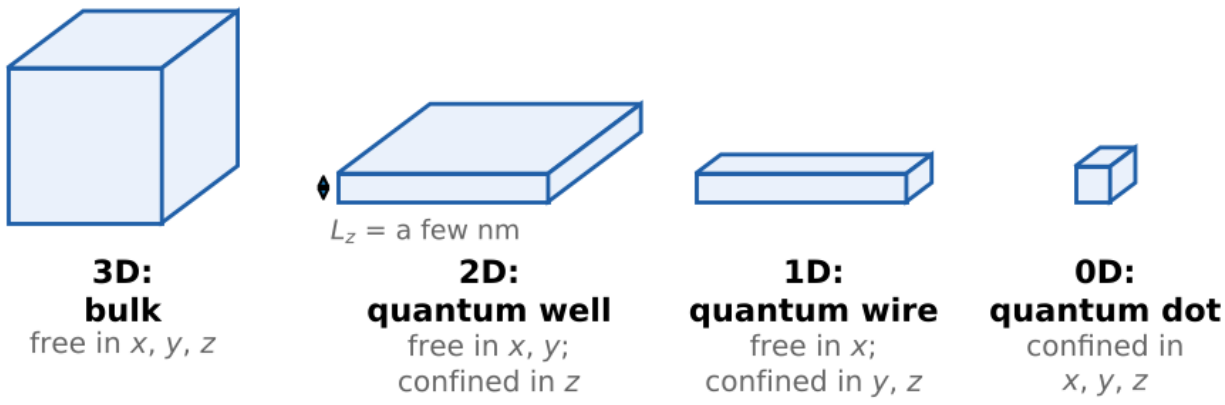


Figure 2: The four structures: bulk (3D), a quantum well (2D), a quantum wire (1D) and a quantum dot (0D)

Reading the figure. In the bulk crystal (left) the carrier is free in all three directions. A quantum well is a thin layer (L_z of a few nm): free in the plane, confined across it. A quantum wire is thin in two directions: free only along its length. A quantum dot is small in all three: no free direction at all, like an artificial atom.

5. Counting the states, case by case

3D (bulk). States with wave number between k and $k + dk$ fill a spherical shell of volume $4\pi k^2\, dk$. One grid point takes a volume $(2\pi/L)^3$ of k-space, so the number of states in the shell, per unit volume of crystal and with 2 for spin, is

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$$N(k)\,dk = 2 \cdot \frac{4\pi k^2\,dk}{(2\pi)^3} = \frac{k^2}{\pi^2}\,dk$$

From $E - E_c = \hbar^2 k^2/2m^*$: $k = \sqrt{2m^*(E-E_c)}/\hbar$ and $dk = \frac{1}{\hbar}\sqrt{\frac{m^*}{2(E-E_c)}}\,dE$. Substituting,

$$D_{3D}(E) = \frac{k^2}{\pi^2}\frac{dk}{dE} = \frac{1}{2\pi^2}\left(\frac{2m^*}{\hbar^2}\right)^{3/2}\sqrt{E-E_c} = \frac{4\pi(2m^*)^{3/2}}{h^3}\sqrt{E-E_c}$$

(Neamen, Eq. 3.72). It is zero at E_c and grows as $\sqrt{E-E_c}$.

2D (quantum well). In one subband the free motion is in the plane: states between k and $k+dk$ fill a ring of area $2\pi k\,dk$, and one grid point takes $(2\pi/L)^2$. Per unit area, with spin,

$$N(k)\,dk = 2 \cdot \frac{2\pi k\,dk}{(2\pi)^2} = \frac{k}{\pi}\,dk$$

Now $E - E_n = \hbar^2 k^2/2m^*$ gives $k\,dk = (m^*/\hbar^2)\,dE$, so

$$D_{2D}(E) = \frac{m^*}{\pi\hbar^2} \quad \text{for } E > E_n$$

a **constant**, the same at every energy. Each new subband adds one more such step, so $D_{2D}(E) = \frac{m^*}{\pi\hbar^2}\sum_n H(E-E_n)$, where H is the unit step. For GaAs ($m^* = 0.067\,m_0$) each step is 2.8×10^{13} cm^{-2}eV^{-1}, and in a 10 nm well $E_1 = 56$ meV, $E_2 = 225$ meV, $E_3 = 505$ meV.

1D (quantum wire). In one subband the free motion is along a line: states between k and $k+dk$ are the two short segments at $+k$ and $-k$, of total length $2\,dk$, and one grid point takes $2\pi/L$. Per unit length, with spin,

$$N(k)\,dk = 2 \cdot \frac{2\,dk}{2\pi} = \frac{2}{\pi}\,dk$$

With $dk/dE = \frac{1}{\hbar}\sqrt{\frac{m^*}{2(E-E_n)}}$:

$$D_{1D}(E) = \frac{2}{\pi\hbar}\sqrt{\frac{m^*}{2(E-E_n)}} = \frac{1}{\pi\hbar}\sqrt{\frac{2m^*}{E-E_n}} \quad \text{for } E > E_n$$

It is **infinite at each subband edge** and falls as $1/\sqrt{E-E_n}$ above it.

0D (quantum dot). There is no free direction, so there is no k to count: only discrete levels E_i, each holding 2 electrons. The density of states is a set of spikes, $D_{0D}(E) = \sum_i 2\,\delta(E-E_i)$.

6. One pattern behind all four

In every case $D(E) \propto (E - E_{edge})^{d/2-1}$, where d is the number of free directions: $d = 3$ gives $\sqrt{E}$, $d = 2$ gives E^0 (a constant), $d = 1$ gives $1/\sqrt{E}$, and $d = 0$ leaves only delta functions. Confinement does not create or destroy states; it gathers them into narrower and narrower ranges of energy. At each E_n the 2D step, divided by L_z, exactly reaches the 3D curve (for GaAs in a 10 nm well both are 2.8×10^{19} cm^{-3}eV^{-1} at E_1), so the staircase hugs the 3D curve from below.

2. Answer

The density of states $D(E)$ counts the allowed electron states per unit energy (and per unit volume, area or length). It comes from counting the allowed k points between k and $k+dk$ and using $E = E_{edge} + \hbar^2 k^2/2m^*$. As free directions are removed by confinement, the shape changes as follows.

- **3D (bulk):** the states lie in a spherical shell $4\pi k^2 dk$, so

$$D_{3D}(E) = \frac{4\pi(2m^*)^{3/2}}{h^3}\sqrt{E-E_c}$$

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It is a smooth parabola-like curve, zero at the band edge and rising as $\sqrt{E}$.

- **2D (quantum well, confined in one direction):** motion across the well is quantised into subbands $E_1, E_2, \ldots$; in each the states lie in a ring $2\pi k\,dk$, so each subband gives a **constant** $m^*/\pi\hbar^2$. $D(E)$ becomes a **staircase**, with a step at each E_n, and has a finite value right at the first edge E_1 (not zero as in 3D). The step corners touch the 3D curve.
- **1D (quantum wire, confined in two directions):** the states lie on a line, so

$$D_{1D}(E) \propto \frac{1}{\sqrt{E - E_n}}$$

for each subband: a **sharp spike** (van Hove singularity) at every subband edge, decaying above it.
- **0D (quantum dot, confined in three directions):** no free motion; only discrete levels, $D(E) = \sum_i 2\delta(E - E_i)$: a set of **delta functions**, like the levels of an atom.

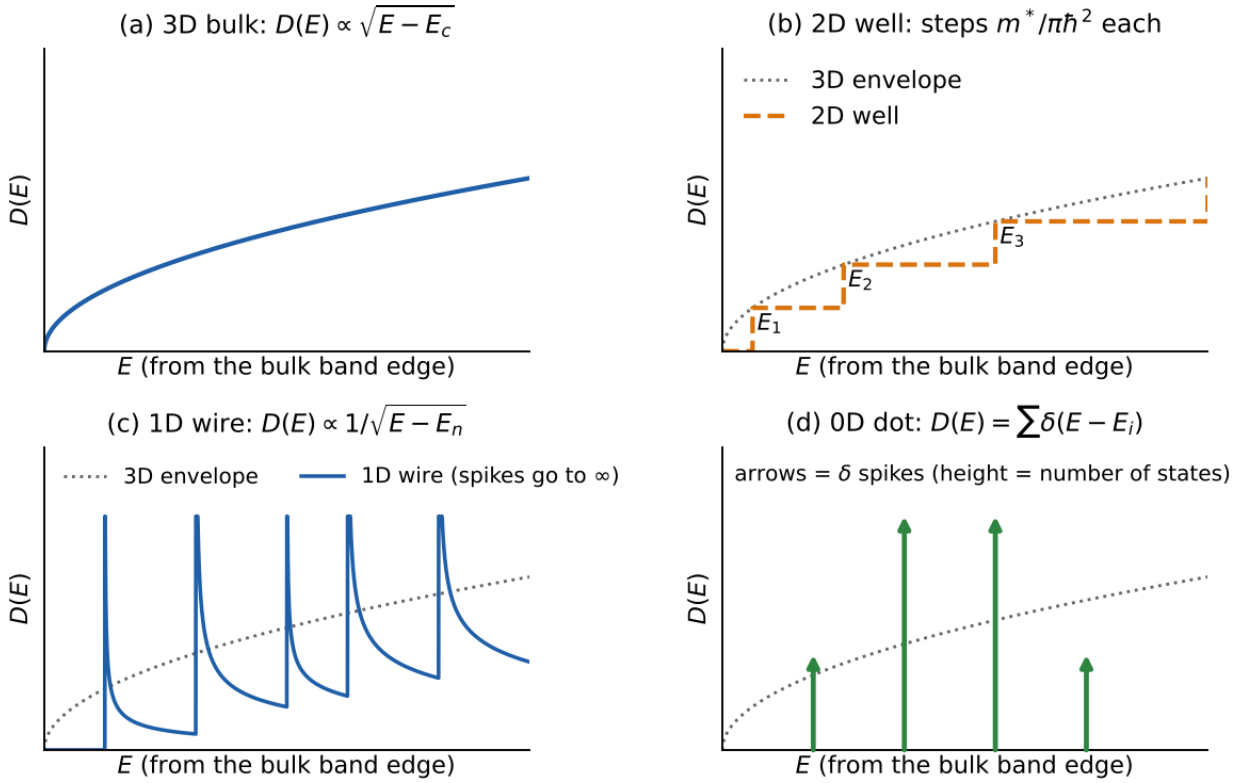


Figure 3: Density of states against energy for 3D, 2D, 1D and 0D, each with the 3D curve dotted grey for comparison (reused from the 2020–21 paper)

Reading the figure. Panel (a) is the smooth solid blue $\sqrt{E}$ curve of the bulk. In (b) the dashed orange staircase of a quantum well rises one step at each subband edge E_1, E_2, E_3 and each corner touches the dotted grey 3D curve. In (c) the solid blue curve of a quantum wire jumps up at each subband edge and decays as $1/\sqrt{E - E_n}$ (the spikes are cut; they really go to infinity). In (d) the green arrows are the delta functions of a quantum dot; a taller arrow means more states at that level.

Qualitative trend: $3D \to 2D \to 1D \to 0D$: smooth $\sqrt{E}$ → steps → spikes → discrete lines. In general $D \propto (E - E_n)^{d/2-1}$ with d the number of free directions. The states are squeezed into narrower energy ranges at the subband edges, and the lowest allowed energy rises above the bulk edge by the confinement energy E_1.

Practical Note: the sharper density of states makes quantum-well and quantum-dot lasers need less current to reach gain and gives them narrower spectra; the 2D staircase also sets the sheet density of the two-dimensional electron gas in a HEMT.

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3. Keyword Explanation

1. Quantum confinement

Squeezing a carrier into a region a few nanometres wide, comparable with its de Broglie wavelength, so that its energy for motion in that direction becomes a set of discrete levels.

2. Subband

All the states that share one quantised level E_n of the confined direction(s) and are free in the other direction(s).

3. Van Hove singularity

A point where the density of states becomes infinite; in a quantum wire it occurs at the bottom of every subband.

4. Notes and Remarks

- **Book:** Neamen derives the 3D density of states (Section 3.4); the 2D, 1D and 0D results follow from the same counting and are the standard low-dimensional results (Sze and Ng).
- With finite barriers the levels E_n are a little lower than the infinite-well values and only a few are bound, but the shapes stay the same.
- The 2D density is per area ($\mathrm{cm}^{-2}\mathrm{eV}^{-1}$) and the 1D per length; divide by the well width (or wire cross-section) before comparing with the 3D value.
- Also asked in 3(b) of the 2020–21 paper (extra reading).

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7.(c) A metal-semiconductor (n-type) junction is formed where the work function of metal is greater than that of the semiconductor. With necessary assumptions, derive expression for electric field in both regions. Also sketch the electric field profile for the both regions. Marks: 9

1. Theory

1. Work function, electron affinity and the vacuum level (Core Idea)

The **vacuum level** E_0 is the energy of an electron just free of the material, at rest. Two quantities describe how deep the electrons sit:

- The **work function** $e\phi$ is the energy needed to take an electron from the Fermi level E_F to the vacuum level. A metal has $e\phi_m$, a semiconductor $e\phi_s$.
- The **electron affinity** $e\chi$ of a semiconductor is the energy from the bottom of its conduction band E_c to the vacuum level. It is a fixed property of the material, independent of doping.

For an n-type semiconductor, $e\phi_s = e\chi + (E_c - E_F)$, and $E_c - E_F = kT\ln(N_c/N_d)$ (Neamen), so ϕ_s depends on the doping.

2. What happens when $\phi_m > \phi_s$?

Before contact, the Fermi level of the n-semiconductor is **higher** than that of the metal (it is $e(\phi_m - \phi_s)$ less deep). When they touch, electrons flow from the semiconductor into the metal, to the lower empty states, until the two Fermi levels line up. This leaves behind, near the surface of the semiconductor, the **ionised donors**, which are positive and fixed: a **depletion region** of width x_n. The electrons that moved sit in the metal as a very thin **negative sheet** right at its surface (a metal has so many free electrons that it needs only an atomic-scale layer to hold this charge). The positive donors and the negative sheet create an electric field pointing from the semiconductor towards the metal, which stops further flow: **equilibrium**.

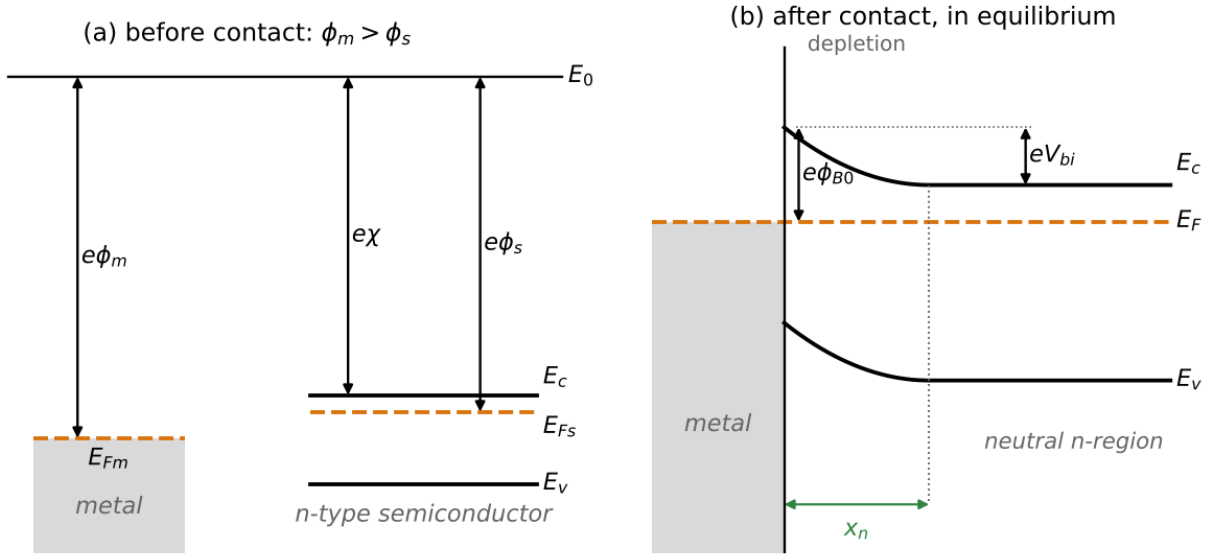


Figure 4: Metal and n-type semiconductor with $\phi_m > \phi_s$: (a) before contact, (b) in equilibrium after contact

Reading the figure. In panel (a) both materials share the vacuum level E_0 (solid black line at the top). The metal's Fermi level E_{Fm} (dashed orange, top of the grey filled region) is $e\phi_m$ below E_0; the semiconductor's E_{Fs} (dashed orange) is $e\phi_s$ below it, higher than E_{Fm}. In panel (b) the Fermi levels are one flat dashed orange line E_F. The semiconductor bands (solid black E_c and E_v) bend *up* towards the metal over the depletion width x_n (green arrow). The barrier seen from the metal is $e\phi_{B0} = e(\phi_m - \chi)$; the barrier seen from the semiconductor is $eV_{bi} = e(\phi_m - \phi_s)$.

3. The tools: Poisson's equation and Gauss's law

The electric field E and the charge density ρ are linked by **Poisson's equation** (one dimension, Neamen):

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$$\frac{dE}{dx} = \frac{\rho(x)}{\epsilon_s}, \qquad E = -\frac{d\phi}{dx}$$

where $\epsilon_s = \epsilon_r \epsilon_0$ is the permittivity of the semiconductor and ϕ the electric potential. In words: the field changes only where there is charge, and its slope is the charge density divided by the permittivity. Across a thin sheet of charge Q' (per unit area) the field jumps by Q'/ϵ (**Gauss's law**). Inside a good conductor in equilibrium the field is zero.

2. Answer

Step 1: Assumptions

1. **Ideal contact:** no interfacial oxide layer and no interface states; the metal and the semiconductor touch directly at $x = 0$ (metal for $x < 0$, semiconductor for $x > 0$).
2. **Uniform doping** N_d in the semiconductor, all donors ionised; $\phi_m > \phi_s$ (rectifying contact).
3. **Depletion approximation:** for $0 < x < x_n$ there are no free electrons, so $\rho = +eN_d$; for $x > x_n$ the semiconductor is neutral ($\rho = 0$). The edge at x_n is sharp.
4. **Metal:** a perfect conductor; its compensating charge is a sheet Q'_m at $x = 0$.
5. **Equilibrium** (zero applied bias), one dimension, constant ϵ_s.

Step 2: The charge density

$$\rho(x) = \begin{cases} Q'_m\,\delta(x) & \text{(metal surface, } x = 0) \\ +eN_d & 0 < x < x_n \\ 0 & x > x_n \end{cases}$$

Overall neutrality: the positive charge per unit area in the depletion region, $eN_d x_n$, equals the negative sheet in the metal, so $Q'_m = -eN_d x_n$.

Step 3: Field in the semiconductor (depletion region)

Poisson's equation with $\rho = eN_d$:

$$\frac{dE}{dx} = \frac{eN_d}{\epsilon_s}$$

Integrate with respect to x (N_d is constant):

$$E(x) = \frac{eN_d}{\epsilon_s} x + C_1$$

Boundary condition: in the neutral region there is no charge and no field, and the field is continuous at x_n (no sheet charge there), so $E(x_n) = 0$:

$$0 = \frac{eN_d}{\epsilon_s} x_n + C_1 \quad \Rightarrow \quad C_1 = -\frac{eN_d x_n}{\epsilon_s}$$

Therefore

$$\boxed{E(x) = -\frac{eN_d}{\epsilon_s}(x_n - x), \qquad 0 \le x \le x_n}$$

(Neamen, Eq. 9.5). The field is negative (it points in the $-x$ direction, from the positive donors towards the negative metal surface), and its magnitude falls **linearly** from the metal to zero at x_n.

Step 4: Field in the neutral semiconductor

For $x > x_n$, $\rho = 0$ gives $dE/dx = 0$, so E is constant; it equals $E(x_n) = 0$:

$$E(x) = 0, \qquad x > x_n$$

Step 5: Field in the metal

Inside a conductor in equilibrium the free electrons would move under any field until it vanished, so

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$$E(x) = 0, \qquad x < 0$$

Check with Gauss's law at $x = 0$: the field jumps from 0 (metal) to $E(0^+)$ (semiconductor). The jump must equal the sheet charge divided by the permittivity:

$$E(0^+) - E(0^-) = \frac{Q'_m}{\epsilon_s} = -\frac{eN_d x_n}{\epsilon_s}$$

which is exactly $E(0)$ from Step 3. So the field is consistent in both regions.

Step 6: Maximum field

The magnitude is largest at the metal surface, $x = 0$:

$$|E_{max}| = \frac{eN_d x_n}{\epsilon_s}$$

Step 7: Depletion width from the potential (to make E_{max} usable)

Integrate $E = -d\phi/dx$, taking $\phi(0) = 0$:

$$\phi(x) = -\int_0^x E\,dx' = \frac{eN_d}{\epsilon_s}\int_0^x (x_n - x')\,dx' = \frac{eN_d}{\epsilon_s}\left(x_n x - \frac{x^2}{2}\right)$$

At $x = x_n$ the potential has risen by the built-in potential V_{bi}:

$$V_{bi} = \phi(x_n) = \frac{eN_d x_n^2}{2\epsilon_s} \quad \Rightarrow \quad x_n = \left[\frac{2\epsilon_s V_{bi}}{eN_d}\right]^{1/2}$$

with $V_{bi} = \phi_m - \phi_s = \phi_{B0} - \phi_n$, $\phi_{B0} = \phi_m - \chi$, $\phi_n = (kT/e)\ln(N_c/N_d)$ (Neamen, Eqs. 9.1–9.7). So

$$|E_{max}| = \frac{eN_d x_n}{\epsilon_s} = \left[\frac{2eN_d V_{bi}}{\epsilon_s}\right]^{1/2}$$

Step 8: The electric field profile

Example values for the sketch

(the question gives none): tungsten ($\phi_m = 4.55$ V) on n-Si ($\chi = 4.01$ V, $N_d = 10^{16}$ cm^{-3}, $N_c = 2.8 \times 10^{19}$ cm^{-3}, $\epsilon_r = 11.7$): $\phi_{B0} = 0.540$ V, $\phi_n = 0.206$ V, $V_{bi} = 0.334$ V, $x_n = 0.208$ µm, $|E_{max}| = 3.22 \times 10^4$ V/cm.

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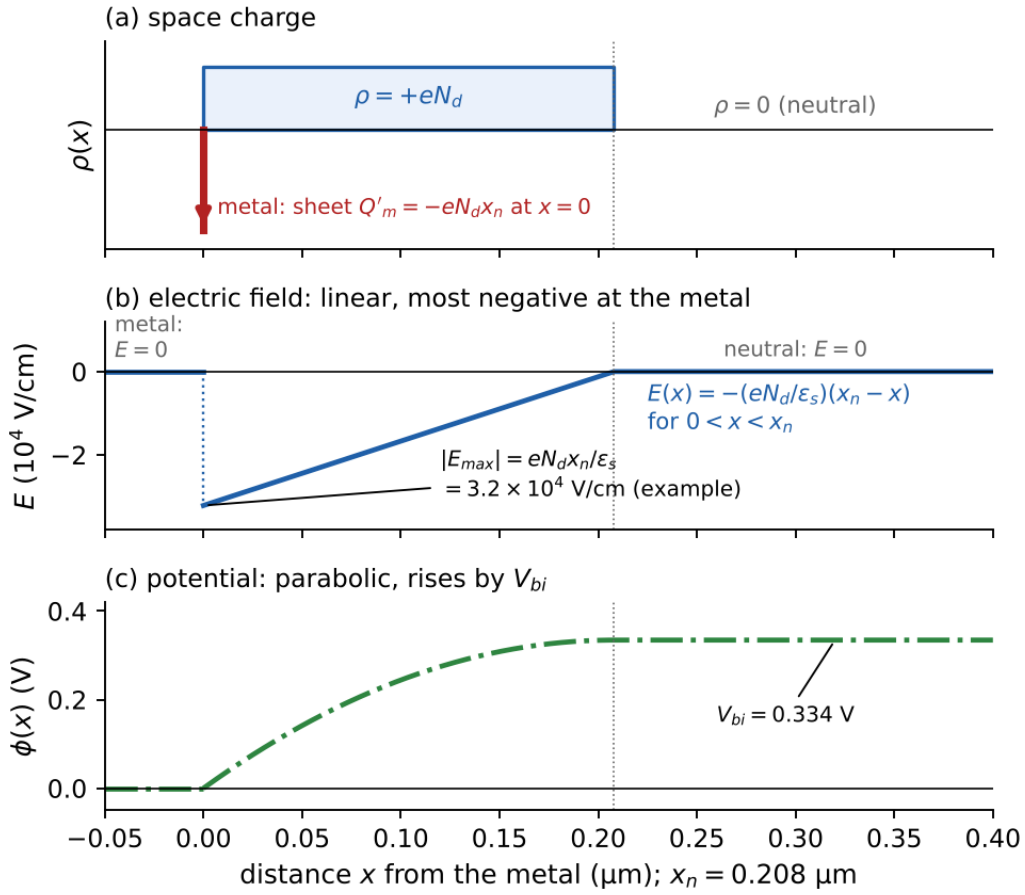


Figure 5: Space charge, electric field and potential of the metal-n-semiconductor junction (example: W on n-Si)

Reading the figure. Panel (a): the shaded blue block is the positive donor charge $+eN_d$ from $x = 0$ to x_n; the thick red arrow at $x = 0$ is the negative sheet charge on the metal surface; beyond x_n there is no charge. Panel (b): the field (solid blue) is zero in the metal ($x < 0$), jumps to $-|E_{max}|$ at the surface, rises **linearly** to zero at x_n, and stays zero in the neutral semiconductor: a triangle. Panel (c): the potential (dash-dot green) rises as a parabola through the depletion region and levels off at V_{bi}.

Summary:

Region	ρ	$E(x)$
Metal, $x < 0$	0 (sheet $-eN_d x_n$ at $x = 0$)	0
Depletion, $0 < x < x_n$	$+eN_d$	$-\dfrac{eN_d}{\epsilon_s}(x_n - x)$
Neutral semiconductor, $x > x_n$	0	0

Table 2: Summary

Steps to remember

Assumptions → charge ($+eN_d$, sheet $-eN_d x_n$) → Poisson, integrate → $E(x_n) = 0$ fixes the constant → $E = 0$ in the neutral region and in the metal → Gauss check at $x = 0$ → $|E_{max}|$ at the metal → $V_{bi} = eN_d x_n^2/2\epsilon_s$ gives x_n.

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3. Keyword Explanation

1. Depletion approximation

Treating the region next to the junction as completely empty of free carriers, with a sharp edge, so that only the fixed dopant ions carry charge there.

2. Built-in potential

The potential difference $V_{bi} = \phi_m - \phi_s$ that develops across the depletion region in equilibrium; it is the barrier electrons in the semiconductor must climb to reach the metal.

3. Schottky barrier height

$\phi_{B0} = \phi_m - \chi$: the barrier seen by electrons in the metal, independent of the doping in the ideal model.

4. Notes and Remarks

- **Reading of "both regions":** the answer gives the field in the metal and in the semiconductor (and, inside the semiconductor, in the depletion and neutral parts). The non-zero part is only the depletion region; the metal holds its charge as a surface sheet, so its field is zero.
- **Sign:** with the metal on the left, E is negative (points to the left, towards the metal). Some books put the metal on the right; then E is positive with the same magnitude.
- **Real contacts:** interface states and image-force lowering change the barrier (see Sze and Ng, Chapter 3), but the triangular field profile stays.
- The same profile is worked with numbers in 3(c) of the 2020–21 paper (extra reading).

5. Final Concise Answer

(a) In $GaAs_{1-x}P_x$ both gaps rise with x: $E_\Gamma = 1.424 + 1.150x + 0.176x^2$ eV and $E_X = 1.900 + 0.144x + 0.211x^2$ eV. They cross at $x_c \approx 0.45$–0.48 ($E_g \approx 2.0$ eV): direct below, indirect above. The **direct (Γ) gap is steeper** (it rises 1.33 eV from GaAs to GaP against 0.36 eV for X), because the Γ minimum is an s-like antibonding state that rises sharply as the smaller P shortens the bonds (large deformation potential), while the p-like X minimum is insensitive.

(b) $D(E)$ counts the states per unit energy. 3D: $D \propto \sqrt{E - E_c}$ (smooth). 2D well: a constant $m^*/\pi\hbar^2$ per subband, a staircase touching the 3D curve. 1D wire: $D \propto 1/\sqrt{E - E_n}$ spikes at each subband edge. 0D dot: delta functions at discrete levels. In general $D \propto (E - E_n)^{d/2-1}$; reducing dimension squeezes the states into sharper features.

(c) With an ideal contact, uniform N_d, the depletion approximation and a perfect-conductor metal: $E = 0$ in the metal, $E(x) = -(eN_d/\epsilon_s)(x_n - x)$ for $0 \le x \le x_n$, $E = 0$ for $x > x_n$; $|E_{max}| = eN_d x_n/\epsilon_s$ at the metal, with $x_n = [2\epsilon_s V_{bi}/eN_d]^{1/2}$ and $V_{bi} = \phi_m - \phi_s$. The profile is a triangle (example: W on n-Si, 3.22×10^4 V/cm, $x_n = 0.208$ µm).

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FORMULA SHEET

EEE459

Compound Semiconductor and Hetero-Junction Devices

Formula Sheet

Session 2021–22 paper

Every formula behind Term Test 1 (Q7); the final will be added

TOPICS

1 Alloy band gaps — 4 formulas

2 Density of states — 8 formulas

3 Metal–semiconductor junction — 8 formulas

Each topic is one table: **Title** · **Equation** · **Notation** · **Reference**.

Prepared by Nur Mohammad

November 2026

BACK OF THE COVER

Before you use the sheet

How to read a row

★ **Direct gap of GaAsP**	$E_\Gamma(x) = 1.424 + 1.150x + 0.176x^2$	E_Γ = gap to the Γ valley (eV) x = P fraction in $GaAs_{1-x}P_x$	7(a)
the name you search for	the formula, exactly as derived	every new symbol with its unit	link: opens the derivation

In the exam

- Write the formula first, then put the numbers in: the formula alone often earns a mark.
- Read the symbols in the Notation column before you substitute, and check the units.
- Not sure where a formula comes from? Its Reference opens the derivation.
- The **Key formulas** page at the end gathers the rows marked ★.
- Lost a symbol? The **Symbol index** at the end links to the row where it first appears.

Contents

1 Alloy band gaps **4**

2 Density of states **4**

3 Metal–semiconductor junction **5**

4 Key formulas **6**

5 Symbol index **7**

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Conventions (Neamen): e is the magnitude of the electron charge, potentials (ϕ, V) are in volts and energies in eV; $kT/e = V_t = 0.0259$ V at 300 K; $\epsilon_s = \epsilon_r \epsilon_0$ with $\epsilon_0 = 8.85 \times 10^{-14}$ F/cm. References point to the solutions of the 2021–22 paper.

Columns: Title · Equation · Notation · Ref. (reference).

1. Alloy band gaps:

Title	Equation	Notation	Ref.
★ **Direct gap of GaAsP**	$E_\Gamma(x) = 1.424 + 1.150x + 0.176x^2$	E_Γ = gap to the Γ valley (eV) x = P fraction in $GaAs_{1-x}P_x$	7(a)
★ **Indirect gap of GaAsP**	$E_X(x) = 1.900 + 0.144x + 0.211x^2$	E_X = gap to the X valleys (eV)	7(a)
Crossover	$E_\Gamma(x_c) = E_X(x_c)$	$x_c \approx 0.45$–0.48, $E_g \approx 2.0$ eV; direct for $x < x_c$	7(a)
Emission wavelength	$\lambda\ (\mu\text{m}) = \dfrac{1.24}{E_g\ (\text{eV})}$	λ = wavelength of a photon of energy E_g	7(a)

Table 1: Alloy band gaps

2. Density of states:

Title	Equation	Notation	Ref.
★ **Parabolic band**	$E = E_c + \dfrac{\hbar^2 k^2}{2m^*}$	k = wave number (m^{-1}) m^* = effective mass (kg) $\hbar$ = reduced Planck constant (J · s)	7(b)
Allowed k in a box	$k = \dfrac{2\pi n}{L}$	L = box size; n = integer; 2 states per k (spin)	7(b)
Infinite-well levels	$E_n = \dfrac{\hbar^2}{2m^*}\left(\dfrac{n\pi}{L_z}\right)^2$	L_z = well width; $n = 1, 2, \ldots$	7(b)
★ **3D (bulk)**	$D_{3D}(E) = \dfrac{4\pi(2m^*)^{3/2}}{h^3}\sqrt{E - E_c}$	states per unit energy per unit volume	7(b)
★ **2D (well)**	$D_{2D}(E) = \dfrac{m^*}{\pi\hbar^2}\sum_n H(E - E_n)$	per unit area; H = unit step; a staircase	7(b)
★ **1D (wire)**	$D_{1D}(E) = \dfrac{1}{\pi\hbar}\sum_n \sqrt{\dfrac{2m^*}{E - E_n}}$	per unit length; a spike at each E_n	7(b)
0D (dot)	$D_{0D}(E) = \sum_i 2\,\delta(E - E_i)$	discrete levels E_i	7(b)
General rule	$D \propto (E - E_n)^{d/2-1}$	d = number of free directions	7(b)

Table 2: Density of states

3. Metal–semiconductor junction:

Title	Equation	Notation	Ref.
★ **Barrier height**	$\phi_{B0} = \phi_m - \chi$	ϕ_m = metal work function (V) χ = electron affinity (V)	7(c)
$E_c - E_F$ in n-type	$\phi_n = \frac{kT}{e} \ln \frac{N_c}{N_d}$	N_c = effective density of states (cm^{-3}) N_d = donor concentration (cm^{-3})	7(c)
★ **Built-in potential**	$V_{bi} = \phi_{B0} - \phi_n = \phi_m - \phi_s$	ϕ_s = semiconductor work function (V)	7(c)
Poisson's equation	$\frac{dE}{dx} = \frac{\rho(x)}{\epsilon_s}$	$\rho = +eN_d$ in $0 < x < x_n$	7(c)
★ **Field, depletion region**	$E(x) = -\frac{eN_d}{\epsilon_s}(x_n - x)$	x from the metal; $E = 0$ in the metal and for $x > x_n$	7(c)
Metal sheet charge	$Q'_m = -eN_d x_n$	C/cm^2, at $x = 0$	7(c)
★ **Depletion width**	$x_n = \left[\frac{2\epsilon_s V_{bi}}{eN_d}\right]^{1/2}$	at zero bias	7(c)
★ **Maximum field**	$\lvert E_{max}\rvert = \frac{eN_d x_n}{\epsilon_s} = \left[\frac{2eN_d V_{bi}}{\epsilon_s}\right]^{1/2}$	at the metal surface	7(c)

Table 3: Metal–semiconductor junction

4. Key formulas:

The formulas marked ★ in the sheet, on one page. Each title links back to its row.

Title	Equation
Direct gap of GaAsP	$E_\Gamma(x) = 1.424 + 1.150x + 0.176x^2$
Indirect gap of GaAsP	$E_X(x) = 1.900 + 0.144x + 0.211x^2$
Parabolic band	$E = E_c + \dfrac{\hbar^2 k^2}{2m^*}$
3D (bulk)	$D_{3D}(E) = \dfrac{4\pi(2m^*)^{3/2}}{h^3}\sqrt{E - E_c}$
2D (well)	$D_{2D}(E) = \dfrac{m^*}{\pi\hbar^2}\sum_n H(E - E_n)$
1D (wire)	$D_{1D}(E) = \dfrac{1}{\pi\hbar}\sum_n \sqrt{\dfrac{2m^*}{E - E_n}}$
Barrier height	$\phi_{B0} = \phi_m - \chi$
Built-in potential	$V_{bi} = \phi_{B0} - \phi_n = \phi_m - \phi_s$
Field, depletion region	$E(x) = -\dfrac{eN_d}{\epsilon_s}(x_n - x)$
Depletion width	$x_n = \left[\dfrac{2\epsilon_s V_{bi}}{eN_d}\right]^{1/2}$
Maximum field	$\lvert E_{max}\rvert = \dfrac{eN_d x_n}{\epsilon_s} = \left[\dfrac{2eN_d V_{bi}}{\epsilon_s}\right]^{1/2}$

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5. Symbol index:

Every symbol the sheet defines, in the order it first appears. The last column links to that row.

Symbol	Meaning	First used in
E_Γ	gap to the Γ valley (eV)	Direct gap of GaAsP
x	P fraction in $GaAs_{1-x}P_x$	Direct gap of GaAsP
E_X	gap to the X valleys (eV)	Indirect gap of GaAsP
λ	wavelength of a photon of energy E_g	Emission wavelength
k	wave number (m^{-1})	Parabolic band
m^*	effective mass (kg)	Parabolic band
$\hbar$	reduced Planck constant (J · s)	Parabolic band
L	box size; n = integer; 2 states per k (spin)	Allowed k in a box
L_z	well width; $n = 1, 2, \ldots$	Infinite-well levels
per unit area; H	unit step; a staircase	2D (well)
d	number of free directions	General rule
ϕ_m	metal work function (V)	Barrier height
χ	electron affinity (V)	Barrier height
N_c	effective density of states (cm^{-3})	$E_c - E_F$ in n-type
N_d	donor concentration (cm^{-3})	$E_c - E_F$ in n-type
ϕ_s	semiconductor work function (V)	Built-in potential

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